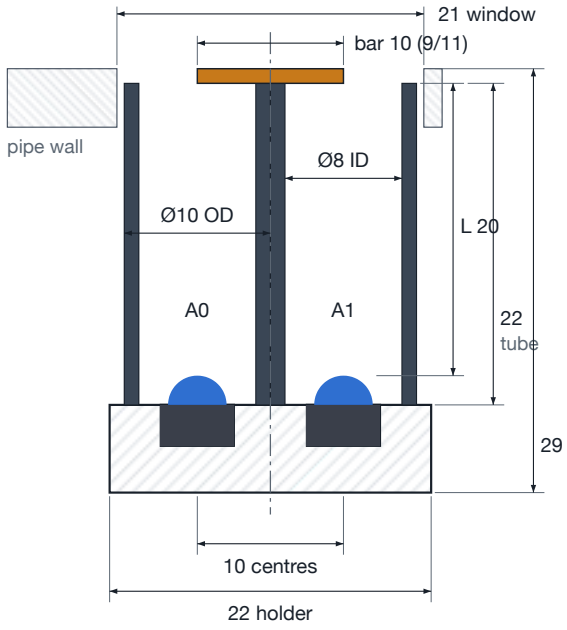
A0/A1 parallel-tube sensor mount

MQ Sentinel · ENGG2000 V3 · Drawing MQS-V3-SNS-002 Rev A (draft) · 2026-10-26 · Prepared by Y. Cui · units mm

Draft for design review. Geometry is calculated with a ray trace (TSSP58P38 lens about 4 × 5.5 mm, beacon at 1 m); it has not been bench-tested. Tolerance ±0.2 unless stated.

1. CAD drawing (8:1)

TOP SECTION · AT LENS HEIGHT

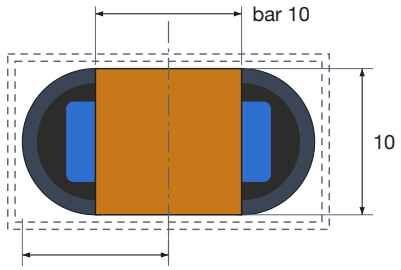


Beacon is toward the top of this view.
Hatched: rear holder (cut). Dark grey: tube walls. Orange: front bar.
L is measured from the lens face (dome front) to the tube opening.
The bar covers the inner half of each opening, so each tube's view leans outward: A0 sees left, A1 sees right.

Critical: bar centred on the centre line within ±0.1 mm.
Tubes parallel to each other and to Datum A within ±0.5°.

VIEW FROM THE BEACON

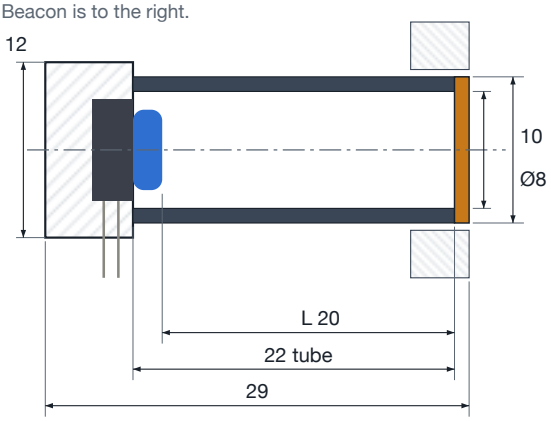
A1 (right sensor) appears on the viewer's left.



Dashed: pipe cut 21 × 11, holder 22 × 12 behind

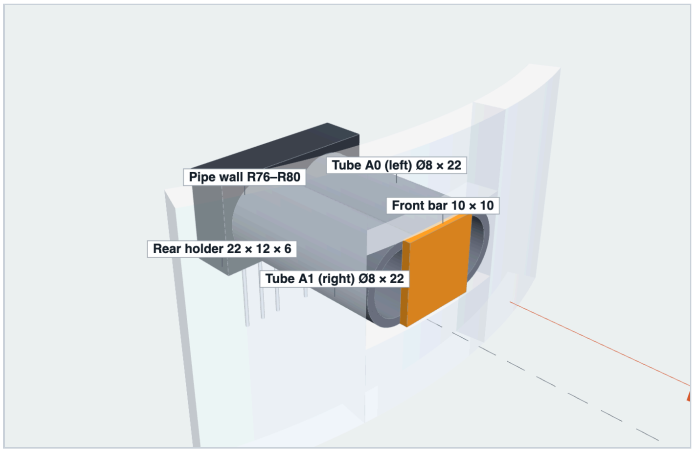
SIDE SECTION · THROUGH A0

Beacon is to the right.

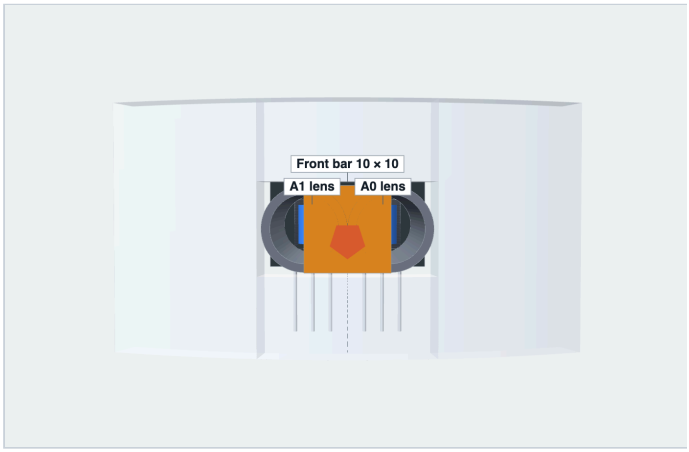


Holder 12 tall (left dimension). Leads exit down through the holder floor.

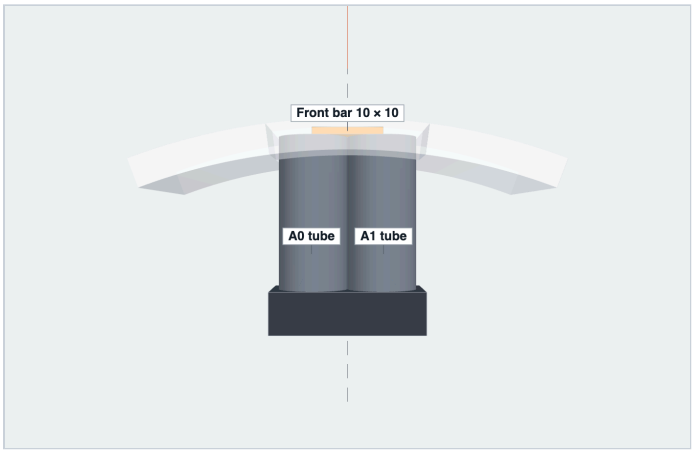
2. 3D views



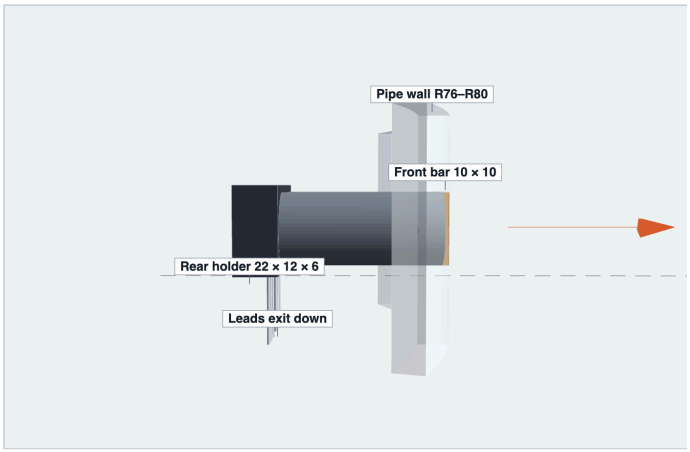
1. Assembled, 3/4 view The mount behind its 21 × 11 window. The bar front is flush with the pipe outside.



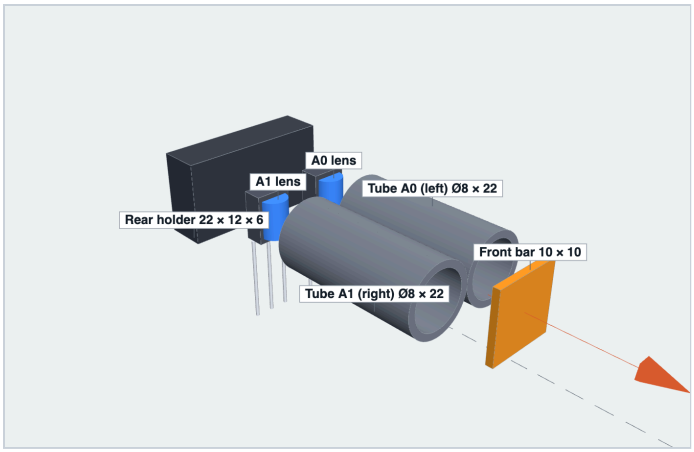
2. From the beacon The 10 mm bar covers the inner half of each tube opening; only the outer half of each lens can see out. A1 appears on the viewer's left.



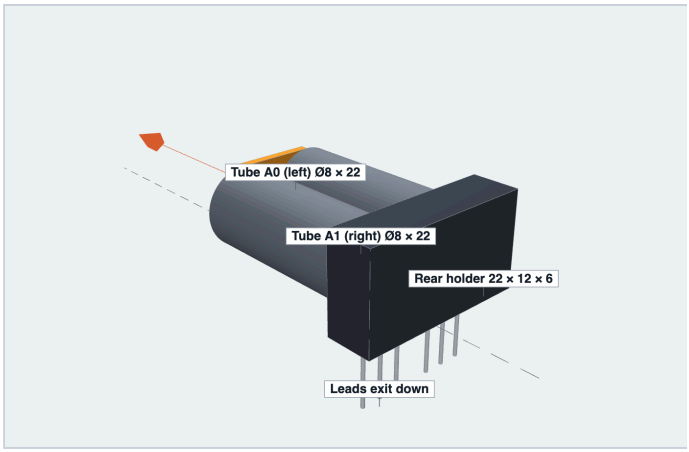
3. Top view (beacon at the top) Two parallel tubes, centres 10 mm apart, on the laser centre line (dashed).



4. Side view Rear holder, tube (22 long), bar at the opening, leads exiting down.



5. Exploded, 3/4 view Assembly order: sensors into the holder, tubes onto the domes, bar across the openings (swap 9 / 10 / 11 mm).



6. Rear 3/4 view From inside the satellite: holder, tube ends and sensor leads.

3. Title block, parts and options

TITLE	PROJECT	DRAWING NO.	REVISION
A0/A1 parallel-tube sensor mount	**MQ Sentinel · ENGG2000 V3**	**MQS-V3-SNS-002**	**A, draft**
DATE	DRAWN	UNITS / TOLERANCE	SCALE
2026-10-26	**Y. Cui**	**mm / ±0.2**	**8:1 (sheet)**

Parts list

#	Part	Size	Qty
1	Rear holder with two sensor pockets and lead holes; mounts to the internal chassis	22 × 12 × 6	1
2	Tube, matte black inside, parallel, butted against the sensor body	Ø8 ID × Ø10 OD × 22	2
3	Front bar, ≤ 1 thick, centred across both openings, swappable	9 / 10 / 11 × 10	3
4, 5	TSSP58P38: A0 (left, pin A0), A1 (right, pin A1); dome inside the tube	5.0 × 6.95 × 4.8	2
6	Pipe window cut, 40 below pipe top, on the laser azimuth	21 × 11	—

Bar width options (ray trace, beacon at 1 m)

Bar	Lit at centre	e at 2.3°	Overlap	Outer limit
9	62%	0.32	±7.0°	±16.6°
10 (start)	50%	0.40	±5.6°	±16.6°
11	38%	0.53	±4.2°	±16.6°

Wider bar: steeper e and less light, but a narrower zone where both sensors see the beacon. Real changes in e will be smaller because the TSSP's response is roughly logarithmic.

Notes

1. The bar's centre must sit on the laser centre line within ±0.1 mm. Each 0.2 mm off centre moves the aim about 0.6°.
2. Both tubes parallel to each other and to the laser direction within ±0.5°.
3. Each lens centred in its tube. Only the dome goes inside; the body sits in the holder pocket.
4. Matte black on every inside surface and on the bar.
5. Direction: beacon on the right makes A1 brighter and e negative, matching the current a1a2_test.
6. Capture zone is ±16.6°; A3–A5 must bring the beacon inside this before fine aim takes over.